

CCDC Linux Quick Reference

Fast triage for services, logs, processes, sessions, and network listeners

Start here: terminal basics

Linux terminal basics

Ubuntu: The Linux command line for beginners

A practical introduction to navigation, files, pipes, and administrator privileges.

Vim keybindings

Vim Keybindings Guide for Beginners

Vim is worth learning: the same navigation keys work in **less** and **journalctl**.

BASIC COMMANDS

ls [path]	List files and directories.
cd <directory>	Change the working directory.
cat <file>	Print a file to the terminal.
less <file>	Read long output one screen at a time.
nano <file>	Edit a text file.
man <command>	Open a command's manual page.

USEFUL PAGER KEYS

j / k	Move down / up.
Space / b	Page forward / backward.
/pattern	Search forward.
n / N	Next / previous match.
q	Quit the pager.

DISCOVER SERVICES

systemctl list-units --type=service	List loaded service units and their current state.
systemctl status <service>	Show status, process details, and recent logs. The .service suffix is optional.
sudo systemctl status <service>	Run with privileges when details are restricted.

CONTROL SERVICES

sudo systemctl start <service>	Start now.
sudo systemctl stop <service>	Stop now.
sudo systemctl restart <service>	Stop and start; commonly used after config changes.
sudo systemctl reload <service>	Ask a supported service to reload without restarting.
sudo systemctl enable --now <service>	Enable at boot and start immediately.
sudo systemctl disable --now <service>	Disable at boot and stop immediately.

INSPECT LOGS

journalctl -xeu <service>

Open detailed logs for one service in a pager. Use arrow or Vim-style navigation keys.

journalctl -f

Follow new journal entries live. Press **Ctrl+C** to stop.

1. Inspect
status

2. Read logs
journalctl

3. Act
restart/stop

4. Verify
status

Processes, Sessions & Network

Find first, verify the target, then contain it.

LIST AND SEARCH PROCESSES

<code>ps -ef less</code>	Show every process in full format, one screen at a time. Focus on PID , PPID , and COMMAND .
<code>ps -ef grep "keyword"</code>	Filter the process list by a command-line keyword.
<code>pgrep -a <name></code>	Find matching PIDs and display command lines.

SEE PARENT-CHILD RELATIONSHIPS

<code>ps -ef --forest</code>	Show the process hierarchy in ps output.
<code>pstree</code>	Display running processes as a compact tree.
<code>pstree -p</code>	Include process IDs in the tree.
Flag note: On Linux ps , the tree option is named --forest , not --tree .	

KILL A PROCESS

<code>kill -9 <PID></code>	Force-kill one process by PID.
<code>pkill -9 <name></code>	Force-kill processes matching a name.
<code>killall -9 <name></code>	Force-kill every process with that executable name.

Target carefully. **-9** sends SIGKILL: the process cannot clean up or save state. In CCDC it can be useful for urgent containment, but preview name matches with `pgrep -a`.

FIND AND REMOVE SHELLS

<code>who</code>	Show logged-in users and their terminals, such as pts/3 .
<code>pkill -9 -t pts/3</code>	Force-kill processes attached to terminal pts/3 .

Check identity and terminal first. Killing the wrong TTY can disconnect a legitimate administrator.

INSPECT LISTENING PORTS

<code>sudo ss -tulpn</code>	Show TCP and UDP listening sockets with owning processes.
Flags: -t TCP -u UDP -l listening -p process -n numeric	

No process names? Re-run with **sudo**. Then identify whether the listener belongs to a service; stop/disable that service, or terminate the process if it should not be running.

FAST TRIAGE SEQUENCE

<code>sudo ss -tulpn</code>	Spot exposed listeners.
<code>ps -ef --forest</code>	Trace the process lineage.
<code>systemctl status <service></code>	Determine whether systemd manages it.
<code>journalctl -xeu <service></code>	Read evidence before changing state.
<code>sudo systemctl stop <service></code>	Contain a service cleanly when possible.
<code>kill -9 <PID></code>	Force containment when necessary.